CHAPTER I
Cybersecurity and the Security Operations Center (SOC)

CSE08 – Elective 08 (IAS / Network Security)

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Lesson Objectives

- Explain why networks and data are attacked
- Identify major cybersecurity dangers and threats
- Describe cybersecurity defenders
- Explain the role of a Security Operations Center (SOC)
- Discuss cybersecurity career paths

Why Cybersecurity Matters

- Most systems are internet-connected
- Data is valuable and vulnerable
- Attacks cause financial and reputational damage
- Cybersecurity protects people and systems

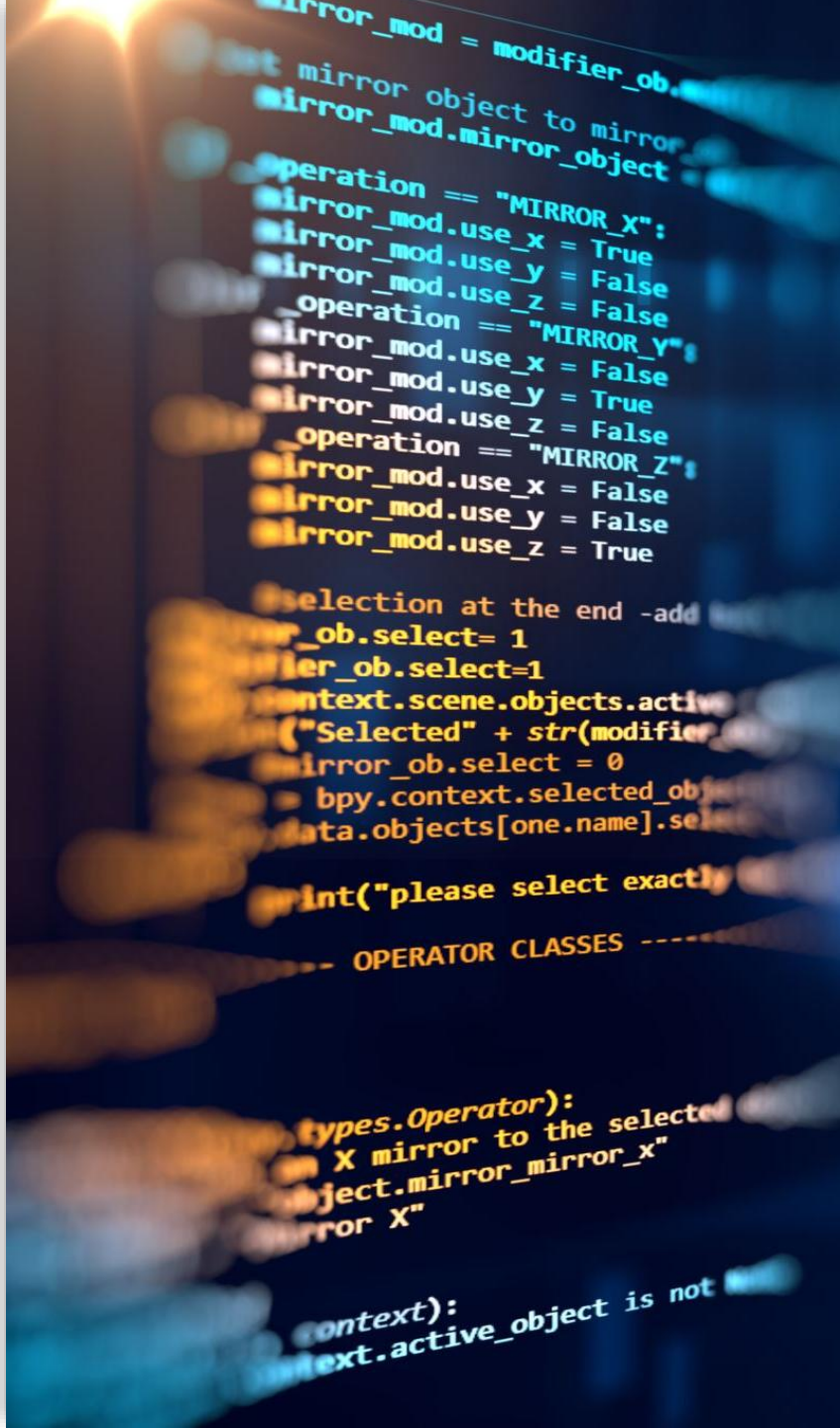
Why Data is Attacked

Common motivations behind cyber attacks:

- Financial gain (fraud, ransomware, scams)
- Espionage (corporate or government spying)
- Disruption (denial of service attacks)
- Revenge or curiosity
- Political or ideological reasons (hacktivism)

Common Cyber Threats

- Malware
- Phishing attacks
- Ransomware
- Unauthorized access
- Data breaches



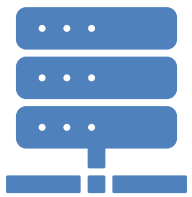
Cybersecurity Defenders

- Security Analyst
- SOC Analyst
- Incident Responder
- Network Security Engineer
- Ethical Hacker (Penetration Tester)

What is a SOC?

A **Security Operations Center (SOC)** is:

- A centralized team
- Responsible for monitoring systems 24/7
- Detects, analyzes, and responds to cyber threats
- Protects an organization’s networks and data



SOC Core Functions

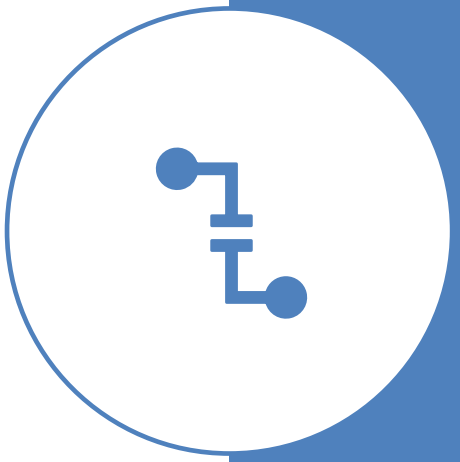
Main responsibilities of a SOC:

1. Monitoring – observing network activity
2. Detection – identifying suspicious behavior
3. Analysis – determining if an attack is real
4. Response – containing and mitigating threats
5. Reporting – documenting incidents

SOC Tools Overview

Common tools used in a SOC:

- SIEM (Security Information and Event Management)
- Firewalls
- Intrusion Detection Systems (IDS)
- Antivirus and endpoint protection tools



Cybersecurity Careers

Common starting roles:

- SOC Analyst
  - (Tier 1)
- Junior Security Analyst
- IT Support
  - Security Track

Skills vs Tools

Tools change.
Threats evolve.
Skills last.

Cybersecurity professionals must be:
Adaptable
Curious
Disciplined
Always learning

Key Takeaways

Cyber threats are constant

Data is valuable

SOC is the heart of cybersecurity